Predicting Employment Outcomes among Young Adults: Education, Well-being, and Machine Learning Evidence from PSID-TAS 2023 & 2021

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1. Introduction

Employment during young adulthood is critical for achieving financial independence, developing work experience, and establishing long-term career success. Difficulties entering the labor market can have lasting economic consequences, including lower earnings and greater financial insecurity.

Educational attainment is widely recognized as an important predictor of employment because it increases skills and labor market opportunities (Yang et al., 2024). However, employment outcomes may also be influenced by psychological well-being and economic circumstances. Individuals with greater confidence and life satisfaction may be better positioned to obtain employment, while financial hardship may create additional barriers to workforce participation (National Center for Education Statistics [NCES], 2024).

Using data from the Panel Study of Income Dynamics Transition into Adulthood Supplement (PSID-TAS), this study examines which characteristics best predict employment outcomes among young adults. The analysis focuses on educational attainment, psychological well-being, and economic hardship using both statistical and machine learning methods.

The central research question is: Which characteristics best predict employment outcomes among young adults? The findings aim to inform policies related to education, workforce development, and economic assistance.

2. Data

2.1 Data Source

This study uses data from the 2023 wave of the Panel Study of Income Dynamics Transition into Adulthood Supplement (PSID-TAS), a nationally representative survey of young adults in the United States. The primary dataset contains 2,434 respondents and 73 variables related to employment, education, family background, financial circumstances, and psychological well-being. Supplementary analyses use information collected in 2021 to predict employment outcomes in 2023.

2.2 Variables

The dependent variable is current employment status, coded as a binary indicator. The primary predictors include educational characteristics (educational attainment, college attendance, educational expectations, and parental education) and a composite well-being score constructed from measures of happiness, life satisfaction, confidence, purpose, and social connectedness. Additional analyses examine economic hardship using SNAP participation as an indicator of financial vulnerability.

2.3 Methods

Employment predictors are evaluated using logistic regression, classification trees (CART), random forests, LASSO, support vector machines (SVM), and double machine learning (DML). The analysis focuses on identifying the characteristics that provide the strongest predictive information about employment outcomes rather than establishing causal relationships.

3. Descriptive Statistics

Table 1 presents summary statistics for the primary variables used in the analysis. The sample consists of 2,434 young adults from the 2023 PSID-TAS survey.

Characteristic	Value
Sample Size	2,434
Employed (%)	57.1
Not Employed (%)	42.9
Mean Well-being Score	4.06
SD Well-being Score	1.06

Table 1. Sample Characteristics

The sample consists of 2,434 young adults from the 2023 PSID-TAS survey. Among respondents, 57.1% were employed and 42.9% were not employed. The average well-being score was 4.06 (SD = 1.17), indicating generally positive levels of well-being. Educational attainment, college attendance, and parental education varied considerably across respondents, providing useful variation for examining predictors of employment outcomes.

4. Comparing Education and Well-being as Predictors of Employment

4.1 Logistic Regression

Logistic regression was first used to estimate the probability of employment:

$$[P(Y_i = 1) = \frac{\exp(X_i\beta)}{1 + \exp(X_i\beta)}]$$

where (Y_i) denotes employment status and (X_i) contains either educational or well-being predictors.

The results reveal a substantial difference between the two domains.
For the education model, several variables were statistically significant predictors of employment. College attendance ($p < 0.001$), current college enrollment ($p < 0.001$), educational aspirations ($p = 0.007$), educational expectations ($p < 0.001$), educational hours ($p < 0.001$), and father's education ($p = 0.003$) all contributed significantly to employment prediction.

In contrast, the well-being model produced considerably weaker results. The composite well-being score was positively associated with employment ($\beta = 0.049$), but the relationship was not statistically significant ($p = 0.205$). The average well-being score among employed respondents was 4.09 compared with 4.03 among non-employed respondents, suggesting only modest differences between the two groups.

These findings suggest that educational attainment provides substantially stronger predictive information than psychological well-being in a traditional statistical framework.

```
Call:
glm(formula = employed ~ ., family = binomial, data = education_tree_data)

Coefficients:
                      Estimate Std. Error z value Pr(>|z|)    
(Intercept)         -0.9846252  0.1816047  -5.422  5.9e-08 ***
HighestEducation     0.0108578  0.0046290   2.346 0.018998 *  
GradeCompleted       0.0112573  0.0108677   1.036 0.300272    
AttendedCollege      0.3564761  0.0342495  10.408  < 2e-16 ***
CurrentCollege       0.4818661  0.0386155  12.479  < 2e-16 ***
EducationAspiration -0.0096882  0.0036026  -2.689 0.007162 ** 
EducationExpectation -0.0728793  0.0204476  -3.564 0.000365 ***
EducationHours      -0.0006538  0.0001731  -3.777 0.000159 ***
MotherEducation     -0.0034145  0.0028676  -1.191 0.233763    
FatherEducation     -0.0035278  0.0012059  -2.925 0.003440 ** 
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Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 3324.3  on 2433  degrees of freedom
Residual deviance: 3046.3  on 2424  degrees of freedom
AIC: 3066.3

Number of Fisher Scoring iterations: 4
```

```
Call:
glm(formula = employed ~ wellbeing_score, family = binomial, 
    data = project_data)

Coefficients:
                Estimate Std. Error z value Pr(>|z|)
(Intercept)      0.09302    0.16184   0.575    0.565
wellbeing_score  0.04898    0.03862   1.268    0.205

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 3317.5  on 2429  degrees of freedom
Residual deviance: 3315.9  on 2428  degrees of freedom
  (4 observations deleted due to missingness)
AIC: 3319.9

Number of Fisher Scoring iterations: 4
```

Figure 1. The logistic regression coefficients for both education and well-being variables.

4.2 Decision Tree Analysis

Decision trees were estimated to identify nonlinear relationships and interactions among predictors.

The education tree generated several meaningful splits based on educational hours, educational expectations, current college enrollment, highest educational attainment, and parental education. The resulting model achieved an out-of-sample classification accuracy of approximately 69.2%.

The well-being tree performed considerably worse, achieving an accuracy of only 55.1%. Furthermore, the tree relied on relatively weak splits involving trust in relationships, responsibility management, and perceptions of others, indicating limited predictive structure within the well-being variables.
The large difference in predictive accuracy suggests that educational variables contain substantially more information about employment outcomes than well-being measures.

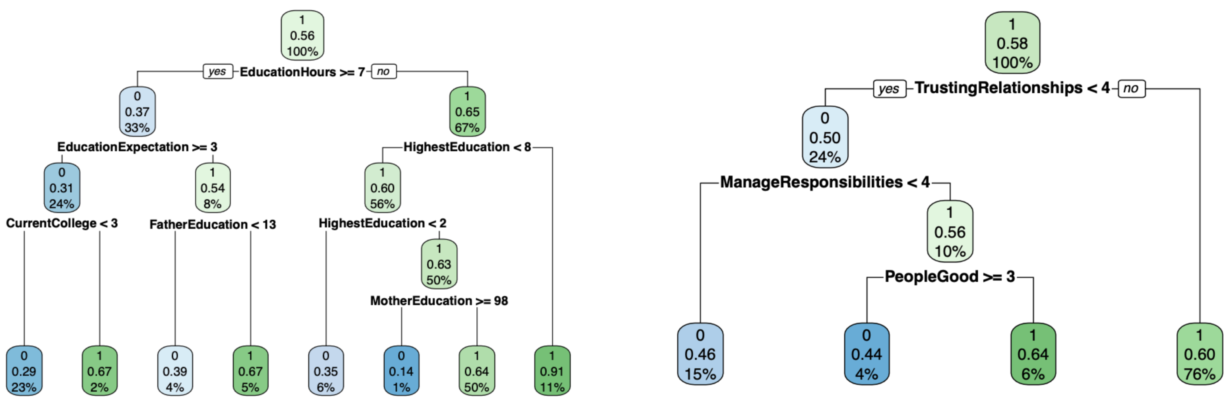

Figure 2. Education and Well-being decision tree

4.3 Random Forest Analysis

Random forests were estimated to evaluate variable importance and improve predictive stability.

For the education model, the most important variables included current college enrollment, highest educational attainment, educational hours, and educational expectations. These variables consistently appeared among the strongest predictors across the forest.

For the well-being model, the most important variables included life satisfaction, responsibility management, trust in relationships, and confidence. However, the overall predictive performance remained weak, with an out-of-bag error rate of approximately 48%.

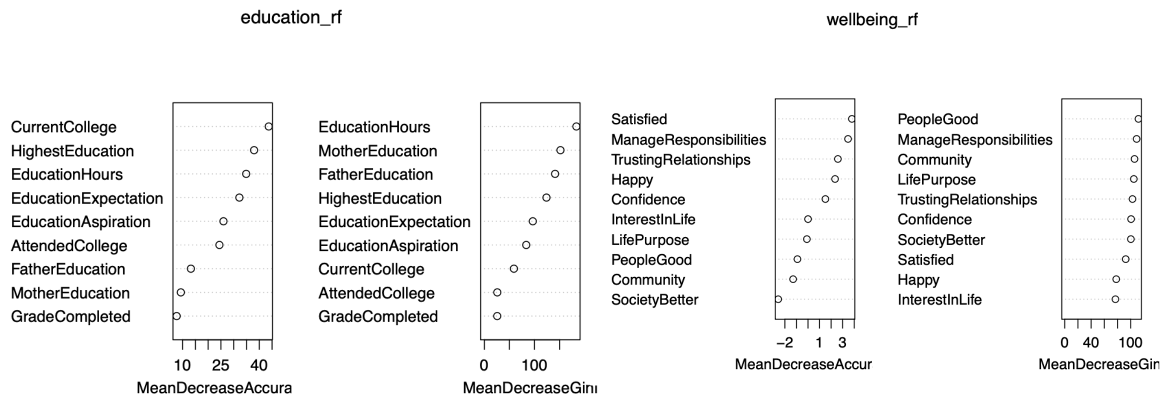

Figure 3. The random forest variable importance plots for both education and well-being models

The random forest results reinforce the conclusion that educational characteristics possess considerably greater predictive power than well-being measures.

4.4 LASSO Variable Selection

To identify the most informative predictors, LASSO regression was estimated using

$$[\min_{\beta} \left[\sum_{i=1}^{n} (y_i - X_i\beta)^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right].]$$

The education model retained multiple predictors after regularization, indicating that several educational variables continued to provide useful information even after penalization.

In contrast, the well-being model performed poorly under regularization. At the one-standard-error rule (λ_{1se}), LASSO eliminated all well-being predictors, suggesting that none contributed substantial predictive information once model complexity was restricted.

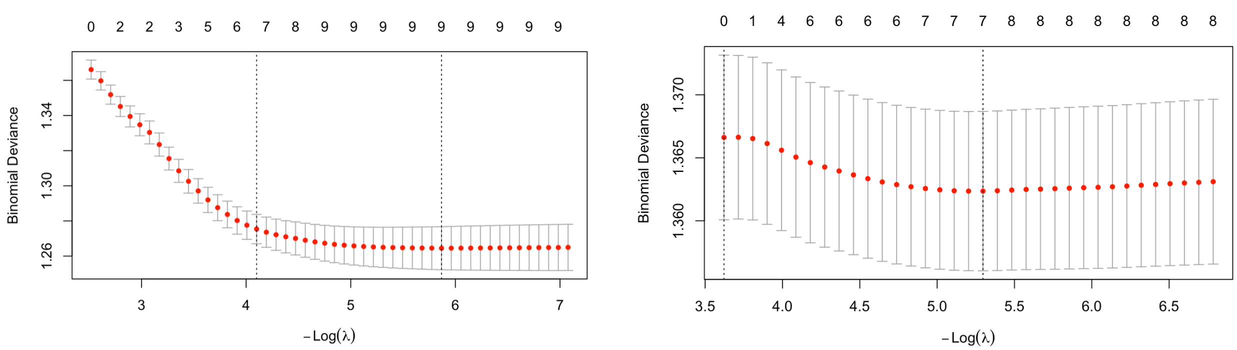

Figure 4. The LASSO coefficient paths for both models

5. Additional Evidence from Economic Hardship and Longitudinal Models

5.1 Food Assistance Participation and Future Employment

To examine whether economic hardship contributes additional predictive information, SNAP participation in 2021 was analyzed as a predictor of unemployment in 2023.

The logistic regression results indicate that SNAP participation was positively associated with future unemployment ($\beta = 0.080$), ($p = 0.017$). Although this effect was smaller than those observed for major educational variables, it remained statistically significant and larger than most well-being measures.

These findings suggest that financial hardship provides useful information about future labor market outcomes and may capture disadvantages not fully reflected by educational attainment alone.

5.2 Double Machine Learning Results

Double Machine Learning (DML) models were estimated using both LASSO and Support Vector Machine learners to identify variables with robust associations with employment outcomes.

Among all variables examined, the strongest positive effect was observed for the belief that young adults should earn their own living, which produced an estimated average treatment effect (ATE) of 0.107. Volunteering behavior also showed a positive association with employment outcomes (ATE = 0.053).

In contrast, emotional, social, and psychological well-being measures generally produced small and statistically insignificant effects. Social anxiety exhibited a weak negative relationship with employment (ATE = -0.020). Overall, the DML results suggest that behavioral characteristics and responsibility-related attitudes provide stronger predictive information than most well-being measures.

5.3 Support Vector Machine Evidence

To further evaluate predictor importance, a Support Vector Machine (SVM) model was estimated using the longitudinal dataset. The SVM results identified previous employment status, years of education, family economic resources, volunteering behavior, and responsibility attitudes as the five most important predictors of employment outcomes.

Educational attainment remained one of the strongest predictors, while well-being measures ranked substantially lower. This pattern is highly consistent with the results obtained from logistic regression, random forests, LASSO, and DML models.

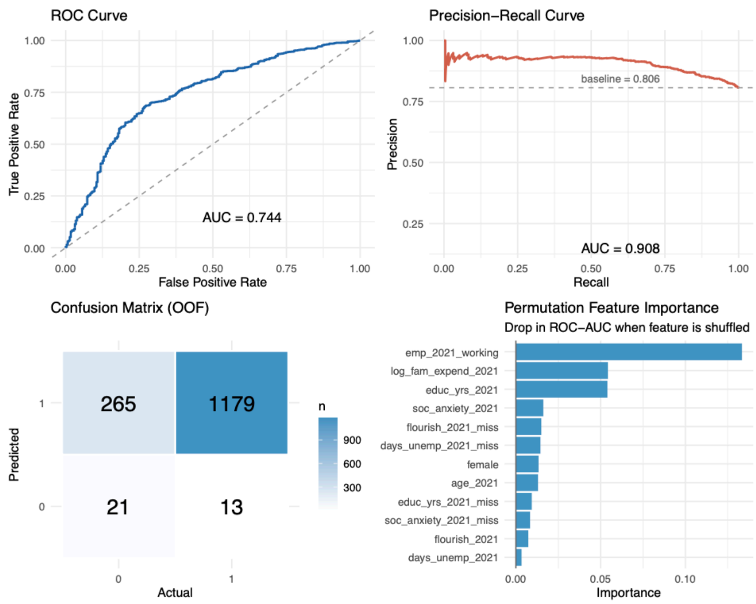

Figure 4. SVM Results. Previous employment status, family economic resources, and years of education were the strongest predictors of employment outcomes. Consistent with earlier analyses, educational and economic factors provided substantially greater predictive power than well-being measures.

6. Discussion

The results show that education is the strongest and most consistent predictor of employment among young adults. Across multiple models, variables such as college attendance, current enrollment, and years of education were repeatedly identified as important factors. This finding highlights the importance of human capital in early labor market outcomes.

In contrast, psychological well-being had weaker predictive value. Although employed individuals reported slightly higher well-being on average, the differences were small and not consistently important across models. This suggests that education and employment-related characteristics provide more useful information for predicting employment.

The supplementary analyses also indicate that economic and social factors matter. SNAP participation was associated with higher future unemployment, while volunteering and responsibility attitudes were positively related to employment. These results suggest that employment outcomes are influenced by both educational background and broader socioeconomic conditions.

From a policy perspective, efforts to improve youth employment should focus on education, skills development, and workforce preparation. Additional support may also be needed for young adults facing economic hardship. Mental health services remain valuable, but they may be most effective when combined with education and employment programs.

This study has several limitations. The analysis is mainly predictive rather than causal, and much of the data is cross-sectional. In addition, some measures are self-reported and may contain errors. Despite these limitations, the findings consistently show that educational and economic factors are more informative predictors of employment than well-being measures alone.

7. Conclusion

This study investigated which factors are most strongly associated with employment outcomes among young adults using data from the PSID Transition into Adulthood Supplement. By comparing results across multiple statistical and machine learning approaches, the analysis found that educational attainment was the most consistent and influential predictor of employment. Indicators such as college attendance, current school enrollment, educational expectations, and total years of education contributed substantially more to predictive performance than measures of psychological well-being.

In addition to education, factors related to economic hardship, volunteering behavior, and personal responsibility showed meaningful associations with employment status. However, well-being indicators generally demonstrated weaker predictive power and were less consistently identified as important across models. These findings suggest that employment outcomes among young adults are shaped more strongly by educational and socioeconomic factors than by self-reported psychological characteristics.

Overall, the results highlight the central role of education in supporting successful transitions into the labor market. Policies and programs that expand educational opportunities, strengthen workforce preparation, and help young adults develop relevant skills may be particularly effective in improving employment prospects and promoting long-term economic stability.

Reference List

National Center for Education Statistics. (2024). *Condition of education: Economic outcomes*. U.S. Department of Education, Institute of Education Sciences. https://nces.ed.gov/programs/coe/indicator/cbc

Yang, X., Chen, Y., & colleagues. (2024). Educational attainment and employment outcomes among young adults. *BMC Public Health, 24*, Article 1892. https://doi.org/10.1186/s12889-024-19266-2